# CHEM 159 — Unit 4 notes

Thermodynamics and equilibrium. A two-page handout for testing the Study Hub uploader.

## Glossary

**Enthalpy** — the heat content of a system at constant pressure
**Entropy** : how spread out the energy of a system is
**Gibbs free energy** :: the energy usable for work at constant pressure
**Catalyst** - a substance that speeds a reaction without being consumed
**Mole** — mol
**Entropy** — second definition, ignored because the first one wins
**Le Chatelier's principle** — equilibrium shifts to oppose the disturbance
**The second law of thermodynamics as stated in the course notes** — total entropy of an isolated system never decreases

## Two lines that will not become terms

**Electronegativity**
the tendency of an atom to attract shared electrons in a covalent bond

## Copied out of a two-column table

| | |
|---|---|
| Molar volume at STP | 22.4 L/mol at 273 K and 1 atm |

## Worked example

The combustion of one mole of methane releases about 890 kJ of energy at 298 K.
A reaction whose Gibbs free energy change is negative is spontaneous at that temperature.
Raising the temperature by 10 K roughly doubles the rate of many room temperature reactions.
A catalyst lowers the activation energy, so more collisions carry enough energy to react.

## A superscript, split onto its own row

One mole of any substance contains $6.022 \times 10^{23}$ particles, which is Avogadro's number.